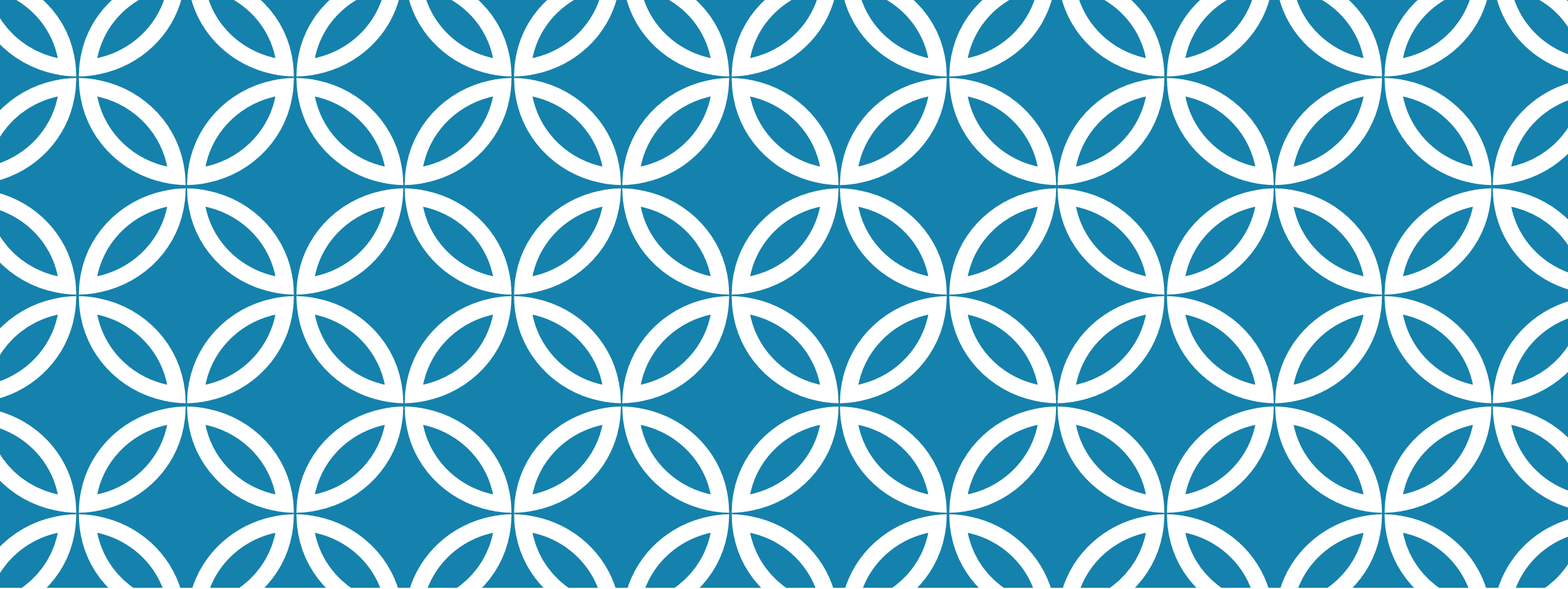




SIMPLIFYING DATABASE MANAGEMENT WITH AMAZON RDS AND AURORA

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AGENDA

What is Amazon RDS?

What are Amazon RDS benefits?

AWS Global Infrastructure

Amazon RDS database instance Multi-AZ deployment(High Availability)

Access Control in Amazon RDS

Introduction to Amazon Aurora

Core concepts and architecture

Demo: Setting up your first Amazon Aurora (MySQL) instance and connecting with Amazon EC2 instance

WHAT IS AMAZON RDS?

A relational database service offered by AWS cloud

It is easy

- to set up
- to operate
- to scale

It is cost-efficient

It supports popular relational database engines

RELATIONAL DATABASE ENGINES

☐ Aurora (MySQL Compatible)



☒ Aurora (PostgreSQL Compatible)



☐ MySQL



☐ PostgreSQL



☐ MariaDB



☐ Oracle

ORACLE®

☐ Microsoft SQL Server



☐ IBM Db2

IBM Db2

WHAT ARE AMAZON RDS BENEFITS?

Manual and automated database backups

Software patching

Detects failure and manages recovery from the failure

Access control using AWS IAM (Identity and Access Management)

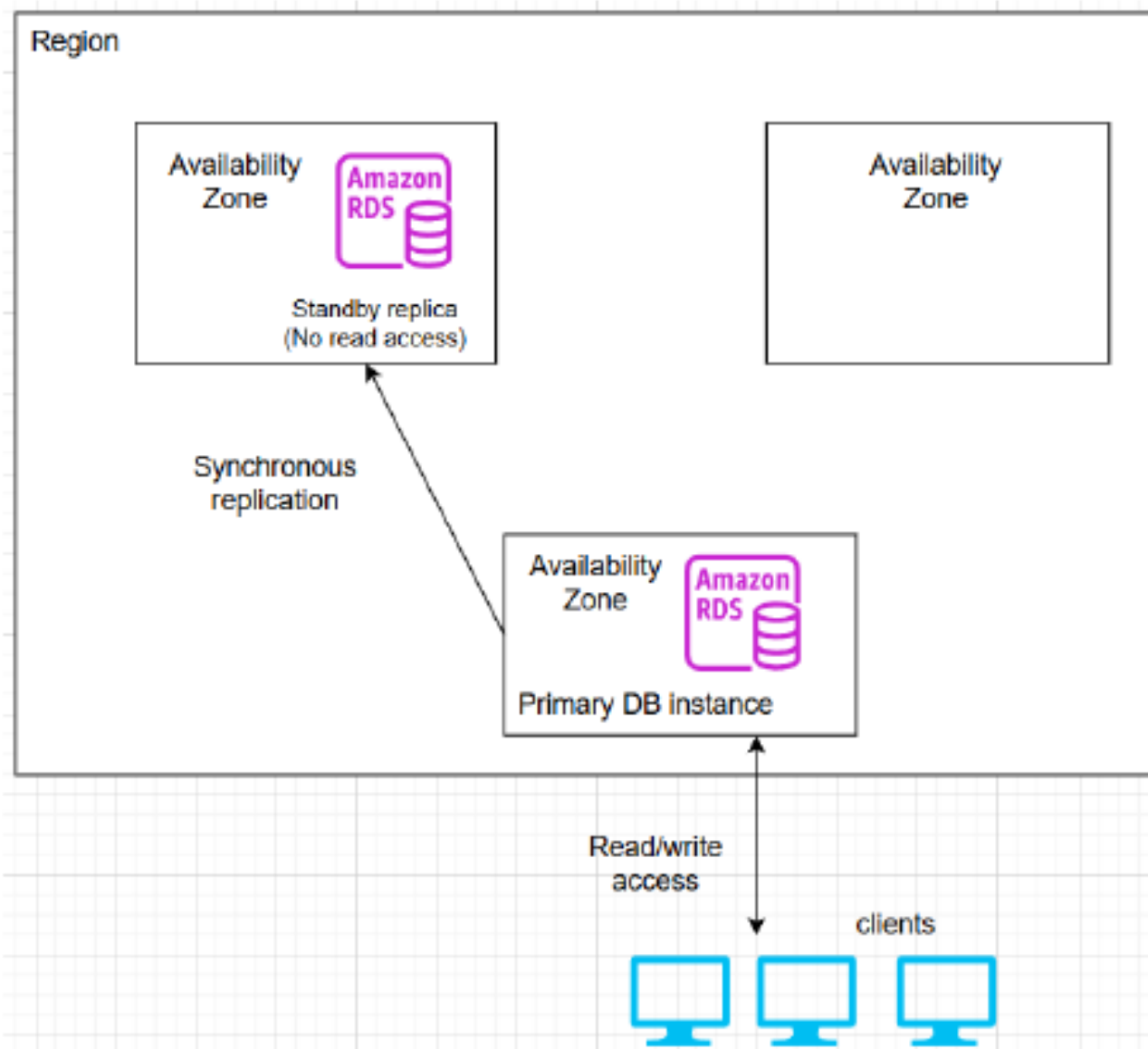


AWS GLOBAL INFRASTRUCTURE

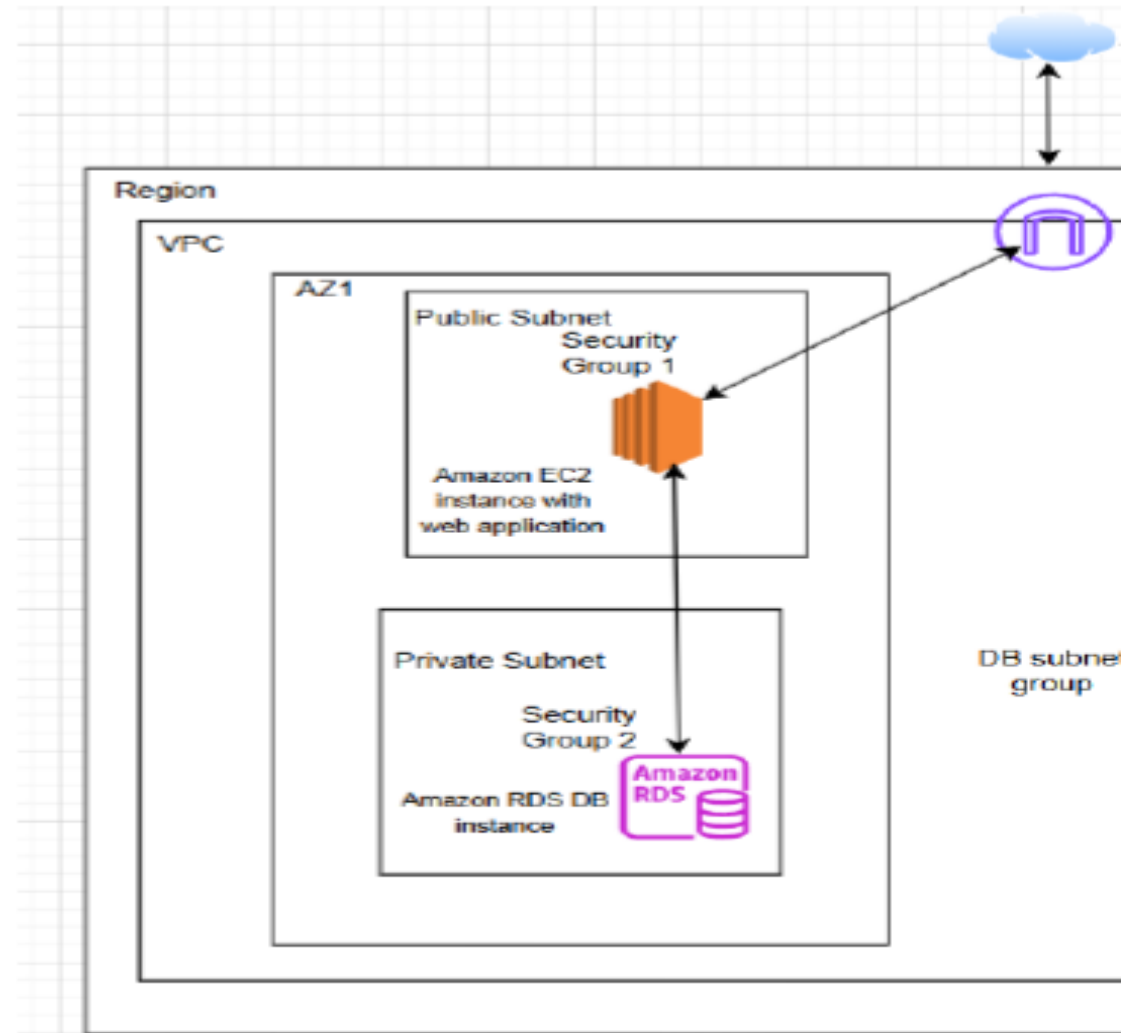
AWS Regions and Availability Zones

HIGH AVAILABILITY

Multi-AZ deployment



ACCESS CONTROL IN AMAZON RDS



WHAT IS AMAZON AURORA?

Amazon Aurora (Aurora) is a fully managed relational database engine

Compatible with MySQL and PostgreSQL.

The code, tools, and applications you use today with your existing MySQL and PostgreSQL databases can be used with Aurora.

Aurora can deliver up to 5 times the throughput of MySQL and up to 3 times the throughput of PostgreSQL

HOW AMAZON AURORA WORKS WITH AMAZON RDS

Choose Aurora MySQL or Aurora PostgreSQL as the DB engine.

Aurora takes advantage of the Amazon RDS features for management and administration.

Aurora uses the Amazon RDS AWS Management Console interface, AWS CLI commands, and API operations to handle routine database tasks such as provisioning, patching, backup, recovery, failure detection, and repair.

AMAZON AURORA DB CLUSTERS

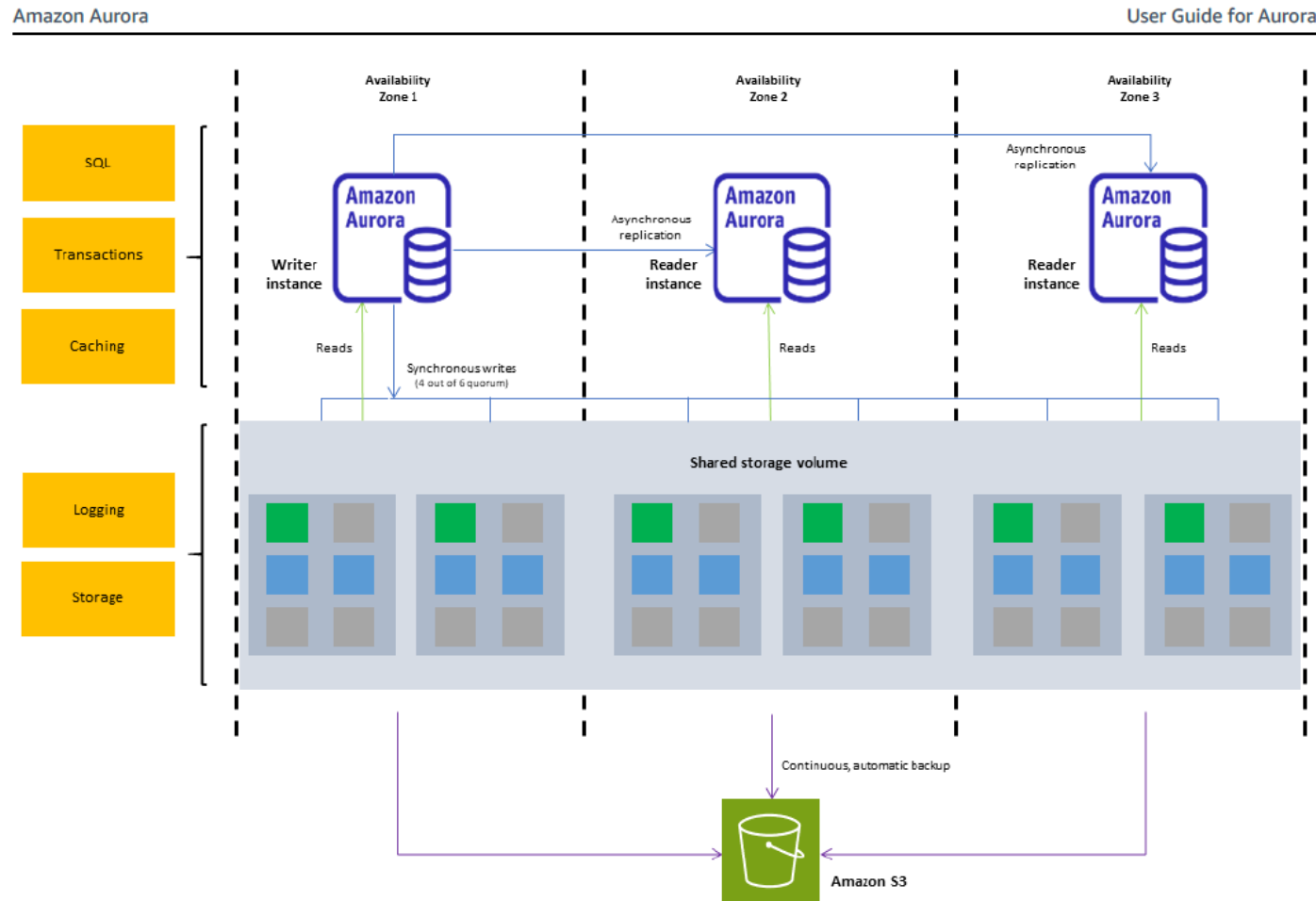
Primary (writer) DB instance – Supports read and write operations, and performs all of the data modifications to the cluster volume.

Each Aurora DB cluster has one primary DB instance.

Aurora Replica (reader DB instance) – Connects to the same storage volume as the primary DB instance but supports only read operations.

Each Aurora DB cluster can have up to 15 Aurora Replicas in addition to the primary DB instance.

AMAZON AURORA DB CLUSTERS CONT..



The relationship between the cluster volume, the writer DB instance, and reader DB instances in an Aurora DB cluster.

DEMO

Setting up your first Amazon RDS Aurora (MySQL) instance and connecting with Amazon EC2 instance

LETS CONNECT



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